

Department of Electrical Engineering

University of Moratuwa

EL2520 – Technical Report Writing for Engineering Studies

Intelligent Robot Vehicle

Technical Project Proposal Report

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1. Objective

To design and develop a LabVIEW-control multifunctional robotic car capable of voice control navigation system and real time environmental adaptability. The car achieves autonomous mobility by using complex sensors and actuators network,

2. Aim

The project aims to achieve the following functional goals:

- To implement a voice command activation system for hands-free vehicle control.
- To implement a voice-based user authentication system that restricts operation to the authorized owner.
- To develop a dual-mode self-parking algorithm capable of detecting and executing both parallel and perpendicular parking slots.
- To provide a post-maneuver alignment score to the user, allowing the system to autonomously redo the parking if precision is insufficient.
- To build a speed control system that automatically slows the vehicle down when it gets dark or visibility is low.
- To deploy a smart detection system that uses behavioral analysis to distinguish between harmless bystanders and potential intruders, while providing user notifications.

3. Scope and Boundaries

The project involves the design, hardware integration, and software development of a LabVIEW-controlled intelligent robotic vehicle. It features voice-based user authentication, autonomous navigation, dual-mode self-parking (parallel and perpendicular) with alignment scoring, light-adaptive speed control, and smart sentry functionality.

Implemented as a small-scale two-wheel-steer (2WS) prototype, its operation depends on sensor accuracy and environmental conditions, with voice commands limited to predefined inputs from the trained user.

4. Literature Review

4.1 Voice Controlled Authentication

Source: *Voice Interaction Recognition Design in Real Life Scenario Mobile Robot Applications* (MDPI, 2023) [1].

Unlike conventional speech to text interfaces that lack user authentication capabilities, this project implements a secure, custom trained machine learning model. By extracting and analyzing, the model generates a precise biometric voiceprint. This ensures that command execution is strictly contingent upon positive owner identification, offering a robust security enhancement over standard voice recognition systems. This means the system will only follow commands if it proves you are the actual owner, making it much more secure than basic voice control.

4.2 Autonomous Self Parking with Parking Alignment Scoring

Source: *PP-US: An Optimized Ultrasonic Sensor Module for Intelligent Parking* (2024) [2].

Basic self parking systems often park inaccurately, because they do not double check their final position. This project fixes that by using ultrasonic sensors to measure exactly how the vehicle is parked. If the LabVIEW software detects that the car is too far from the edge or parked at an angle, it automatically makes the vehicle pull out and try parking again.

4.3 Intruder Detection and Behavior Analysis

Source: *Passive Infrared Sensor-Based Occupancy Monitoring: A Review* (2024) [3].

Normal motion sensors trigger false alarms too easily from pets or passing breezes. We fix this by combining a thermal motion sensor with an ultrasonic sensor for distance tracking. The LabVIEW software measures how long something actually stays in the area, ignoring quick, harmless movements and only sending an alert if someone is suspicious.

4.4 Adaptive Speed Limiting via Light Variation

Most small robotic cars run at a single, fixed speed regardless of their surroundings. This project takes a smarter approach. By using a custom light sensor (LDR) circuit, the system constantly checks the ambient light. When it gets dark and visibility drops, the LabVIEW software automatically slows the motors down by reducing their PWM signal, acting just like a real car driving carefully at night,

5. Estimated Budget

Table 1 presents the estimated component costs for building the prototype.

Table 1: Estimated Component Budget

Component	Quantity	Price (Rs.)
Ultrasonic Sensor(HC-SR04)	4	750
PIR Sensor (HC-SR501)	1	230
LDR	1	90
Motor Driver (L298N)	1	760
Li-ion Battery	2	1300
Robot Car Chassis, Wheels, Motors	1 set	2000
Servo Motors (SG90)	2	600
Encoder Motors	2	3400
Total		Rs. 9130

6. Methodology (Logic and Implementation)

The system operates as a multi stage process controlled by LabVIEW communicating with an Arduino micro-controller over a DAQ/serial interface. Figure 1 presents the complete operational flowchart of the system.

6.1 Stage 1: System Initialization

When the system boots up, LabVIEW immediately configures all essential hardware connections, including the Arduino DAQ interface, audio input, LDR analog line and the required digital pins for the PIR and ultrasonic sensors. Simultaneously, the LDR initiates a continuous background loop to dynamically regulate the maximum allowable PWM limit.

6.2 Stage 2: Sensor Data Acquisition

Three sensor streams are activated simultaneously:

- The microphone continuously captures audio for voice processing.
- The custom LDR circuit reads ambient lux levels.
- The ultrasonic and PIR sensors await their respective intruder conditions.

6.3 Branch A: Voice Command and AI based Authentication

All voice input is first passed through the AI based authentication module. MFCCs are extracted from the captured audio and fed to the trained classification machine learning model. If the voiceprint does not match the registered owner, the audio is discarded and access is denied. Upon successful authentication, the decoded command is routed to one of three operational branches.

6.4 Branch B: Smart Sentry — Intruder Behavior Analysis

1. The PIR sensor monitors for thermal motion signatures.
2. Upon a PIR trigger, ultrasonic targeting is activated to track the position of the subject.
3. By tracking the duration of an individual's presence, the LabVIEW software effectively filters out brief movements to isolate genuine security risks.
4. If the that time exceeds a predefined threshold, a verified alert is triggered, connecting to the IoT cloud module and pushing a notification to the user's phone through telegram.

6.5 Branch C: Standard Navigation with Adaptive Speed Control

1. The decoded voice command identifies the directional instruction (Move forward, reverse, left, right, stop, System Armed).
2. The current LDR reading is applied to dynamically cap the L298N motor driver's maximum PWM output.
3. The L298N motor direction registers are set accordingly.
4. The vehicle drives until a "Stop" command is received.

6.6 Branch D: Autonomous Self-Parking with Alignment Scoring

1. Ultrasonic sensors perform a lateral side-scan to measure available parking space dimensions.
2. LabVIEW classifies the space as suitable for parallel or perpendicular parking.
3. The appropriate sequence of steering servo and motor actuation commands is executed to complete the parking process.
4. Post-parking, ultrasonic sensors measure the vehicle's offset from the space boundaries.

LabVIEW computes an alignment score and outputs it to the user. If the score falls below the minimum precision threshold, a re-parking sequence is automatically initiated.

6.7 Stage 3: Hardware Actuation

All branches converge at the hardware actuation stage, where LabVIEW sends the final PWM and direction signals to the L298N motor driver (for drive motors) and pulse-width commands to the SG90 steering servos.

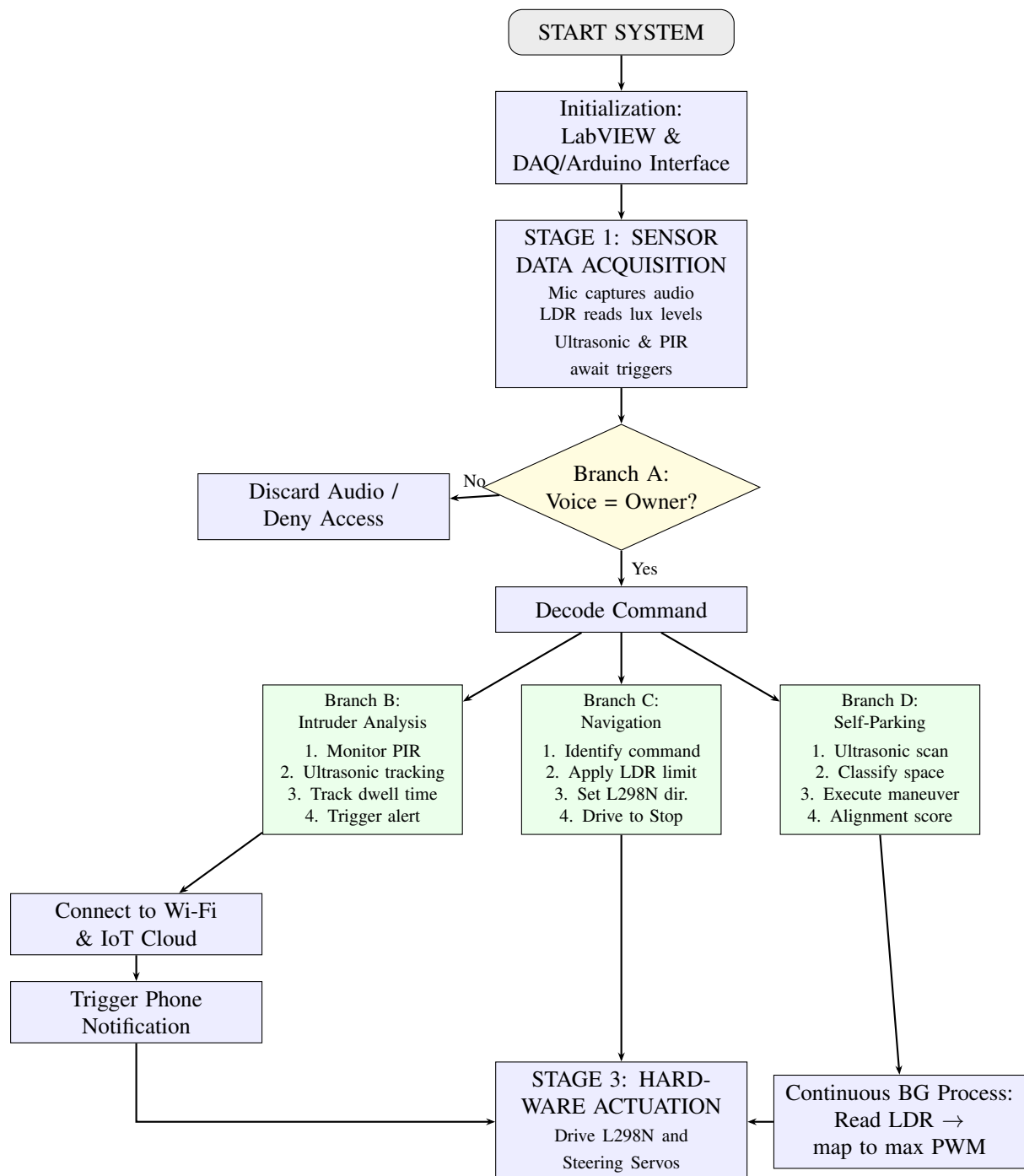


Figure 1: System Operational Flowchart — Intelligent Robotic Vehicle

6.8 LabVIEW Implementation

Figure 2 shows the image of the intelligent robotic vehicle with all sensors and actuators.

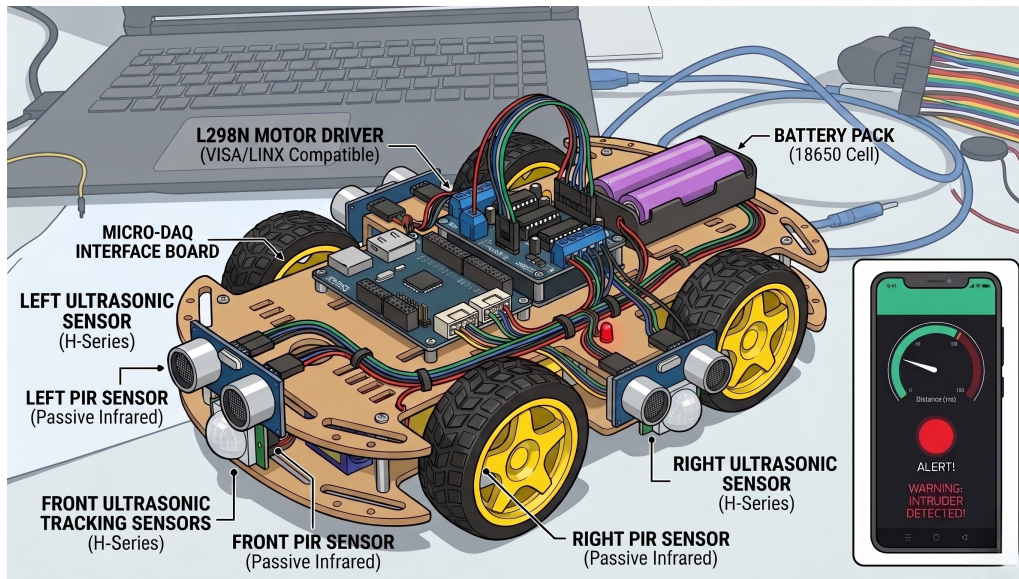


Figure 2: Intelligent Robotic Vehicle

Figure 3 presents the LabVIEW Front Panel illustrating the data flow displaying real time sensor readings, authentication status, and vehicle control interface.

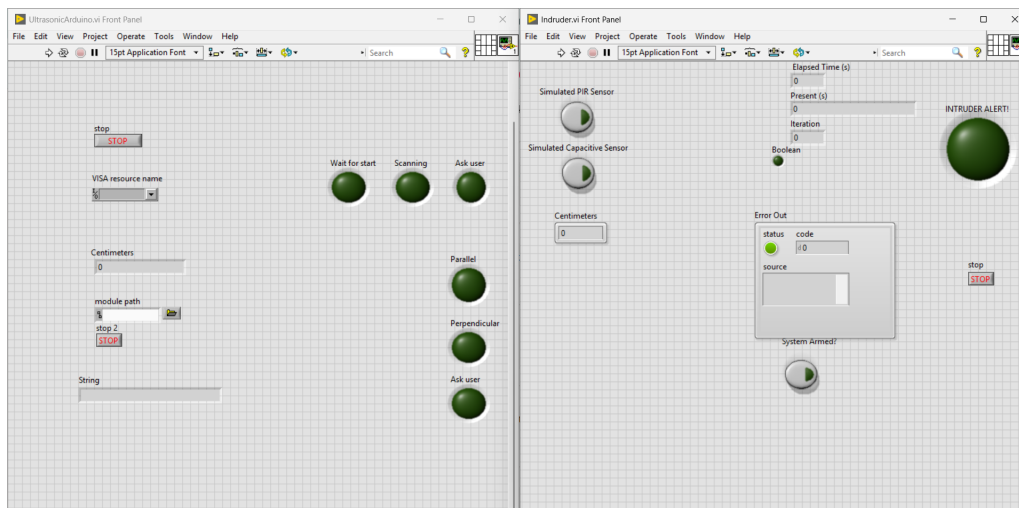


Figure 3: LabVIEW Front Panel

7. Conclusion

This prototype establishes a scalable framework for next generation autonomous vehicles by integrating LabVIEW and Arduino for real-time sensor processing. The vehicle features a suite of advanced capabilities, including voice-verified access, environmentally adaptive speed regulation, behavioral security tracking, and automated parking alignment. By applying targeted sensor fusion and adaptive control techniques, the system effectively neutralizes real-world vulnerabilities like false intrusion triggers and blind parking errors. Its modular design not only facilitates streamlined testing but also provides a clear pathway for deployment in larger, commercial autonomous systems.

References

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